

# Refusal Is Not Harm

*What I set out to measure, what I found instead, and why it matters*

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I set out to test whether compressing a language model makes it easier to jailbreak. It does not. But in working out *why* that answer was so clean, I found that the yardstick the whole field uses to score these attacks ranks them in the wrong order.

## 1 The question I started with

Large language models are expensive to run, so people shrink them. Quantization is the standard trick: store each number in the model using 4 bits instead of 16. The model gets roughly four times smaller and runs on cheaper hardware.

The obvious worry is that the safety training gets damaged in the process. A shrunk model might answer questions the original would have refused. My internship was to test that, specifically for *multi-turn* attacks, where the attacker does not ask the dangerous question directly but works up to it over several messages.

### IN PLAIN TERMS

Does squeezing a model to make it cheaper also make it easier to trick?

## 2 The answer, and the problem it exposed

Compression barely moved the needle. Across the comparisons I ran, the shrunk models refused about as often as the originals.

But something did not add up. The standard scoring method said these attacks were succeeding **83% of the time**. If the attacks work that well, and compression does not change that, then either the models have no safety at all, or the score is measuring the wrong thing.

So I checked what the models were actually saying.

## 3 Finding one: the scoreboard is upside down

Nearly every paper in this area scores an attack as successful when the model *fails to say no*. That is the entire test. Nobody checks whether the answer was actually dangerous.

I added two things the field leaves out:

- **A harm judge.** A separate safety model read all 6,000 responses and labelled whether each one was genuinely harmful.
- **A control condition.** I asked every model the same final question *without* the attack conversation in front of it. If it answers that freely anyway, the attack deserves no credit.

With those two additions, an attack only counts if the model refused the question cold, did not refuse it after the attack, *and* the answer was actually harmful.

The result reverses the ranking of the two attacks I tested:

| ATTACK                         | STANDARD SCORE | HARM IT ACTUALLY CAUSED |
|--------------------------------|----------------|-------------------------|
| CoSafe (indirect reference)    | 82.9%          | 1.0%                    |
| Crescendo (gradual escalation) | 23.8%          | 7.0%                    |

11 matched model and precision configurations. Both differences are statistically significant ( $p < 0.001$  and  $p = 0.02$ ) and they point in opposite directions.

The attack that looks three times more dangerous is seven times less dangerous. Of everything CoSafe scores as a win, **93% contains nothing harmful at all**. It is credited for answers the model was always willing to give.

#### IN PLAIN TERMS

Imagine grading a lock-picking test by counting how often the door opens, without noticing that most of those doors were never locked. You end up praising the wrong burglar.

## 4 Finding two: the risk sits in the model family, not the compression

Once I could measure harm properly, three clearly different behaviours appeared, and none of them had anything to do with compression.

| MODEL      | BEHAVIOUR             | WHAT IT MEANS                                                                                                          |
|------------|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| Llama-3.x  | Immune, and hardens   | Indirect attacks do nothing. Escalation makes it <i>more</i> cautious, not less (refusal rises 23 points on the 3B).   |
| Qwen2.5    | Genuinely vulnerable  | Refuses <b>99%</b> of harmful questions asked directly, but only <b>7.5%</b> once a conversation is built around them. |
| Mistral-7B | Barely safety-trained | Refuses only 18.5% of anything. Its high attack score is real, not an artifact.                                        |

The Qwen result is the important one. Judged by the single-question tests everyone runs, Qwen looks like the safest model in the study. It is the least safe. That gap is invisible to the benchmarks currently in use.

Mistral matters too, for the opposite reason: it is the case where the standard score is *correct*. That rules out the possibility that my harm judge is just insensitive. The metric works fine on models that do not refuse much. It breaks specifically on safety-trained models, which are the only ones anyone actually benchmarks.

## 5 Finding three: compression matters only when the attack adapts

The null result turned out to be conditional rather than universal, and the split is sharp.

| ATTACK TYPE                                         | EFFECT OF COMPRESSION | SIGNIFICANT?                           |
|-----------------------------------------------------|-----------------------|----------------------------------------|
| Fixed script (attacker never adapts)                | 1.4 points            | 0 of 7 comparisons                     |
| Adaptive (attacker quotes the model back to itself) | <b>8.6 points</b>     | <b>3 comparisons, after correction</b> |

The cleanest evidence is a single model, Qwen-3B, under one single change from 16-bit to 4-bit, tested three ways:

| ATTACK                  | CHANGE FROM COMPRESSION | P-VALUE             |
|-------------------------|-------------------------|---------------------|
| One direct question     | +0.5 points             | 1.00                |
| Fixed multi-turn script | +1.0 points             | 0.77                |
| Adaptive escalation     | <b>+16.5 points</b>     | <b>&lt;0.000001</b> |

Same model, same compression, three attacks. Only one moves. The only thing that differs is whether the attack's later messages depend on what the model said earlier.

#### WHY THIS HAPPENS

An adaptive attack quotes the model's own words back at it. Change the model even slightly and you change what it says, which changes the attack, which changes what happens next. The differences compound. You are no longer comparing two models against one attack, you are comparing two different attacks. That is a measurement problem, not a safety problem, and it applies to anyone comparing model variants with adaptive attacks.

## 6 Two bugs I found in my own work

I am listing these because catching them is the part I am most confident about.

### The held-out test set was not held out

The dataset I validated against became access-restricted partway through. The code quietly fell back to a small built-in list of 34 prompts, repeated four times. So every prompt in my supposedly unseen test set had also been used for training. The reported 100% accuracy was meaningless. I found it while debugging something unrelated, switched to a proper 520-prompt copy stored in the repository, and re-ran with a genuine held-out quarter. The numbers came out similar, but now they mean what they claim.

### Silent crash on multi-GPU machines

One analysis step assumed the whole model sat on a single GPU. On machines that split the model across two, it crashed. A safety check downstream then correctly refused to continue, but it reported the wrong reason, so the real cause was hidden. Fixed.

### A limitation I am still carrying

One diagnostic step uses only 16 prompts and is unstable, picking slightly different answers on repeat runs. It affects a supporting measurement only, not the headline refusal or harm results, which come from actual generated text. The fix is cheap and I have not run it yet.

## 7 Scale, cost, and what went wrong operationally

| ITEM                           | VALUE                                    |
|--------------------------------|------------------------------------------|
| Models tested                  | 5 families                               |
| Compression formats            | 4 (16-bit, NF4, AWQ, GPTQ)               |
| Attacks                        | 2 multi-turn, plus a single-turn control |
| Responses generated and judged | 6,000                                    |
| Compute environments           | 3 (laptop, free Kaggle GPU, rented A100) |
| Total cloud spend              | <b>around \$12</b>                       |

Cost stayed low by putting the uncompressed runs on hardware I already had, using free quota for the 6,000-response judging pass, and renting a serious GPU only for the parts that genuinely required it.

Things that broke, and what they taught me:

- One compression format silently requires a newer GPU generation than I was using. The library retried the build for hours instead of reporting the failure. Cost about four and a half hours. The runner now checks the hardware once and skips it.
- Redirecting a cache directory also moved my login credentials, so downloads started failing an hour into a run.
- A storage fault on a network drive destroyed one run's output.
- The laptop run died from memory pressure with two large models loaded.

The generalisable fix: I now judge results before generating more, so a session that dies later still keeps its labels, and every step is restartable, so a lost session costs one step rather than the whole run.

### 8 What this changes for other people

The recommendation is small and cheap. Alongside the usual score, report what the model would have answered *without* the attack. That is one extra generation per test case, and it makes the inflation visible immediately.

Without it, a benchmark can rank a harmless attack above a dangerous one, and the people relying on that ranking to decide what to defend against will defend against the wrong thing.

### 9 What I would do next, in order

- **A second harm judge, plus human checking on a sample.** Right now one model's judgement carries the central claim. This is the biggest gap and it needs no new GPU time.
- **A third, fully adaptive attack.** My explanation predicts the compression effect should get *larger* as the attacker adapts more. Testing that gives a dose-response curve, which is much stronger evidence than what I have. It also puts the ranking claim on three points instead of two.
- **Fill two missing cells.** Compressed Mistral and one Llama-8B configuration were never collected because compute ran out.
- **Re-test published rankings.** Check whether adding the control condition changes the conclusions of existing multi-turn benchmarks.

### 10 Numbers worth remembering

| MEASURE                                             | VALUE                    |
|-----------------------------------------------------|--------------------------|
| Standard score, CoSafe vs Crescendo                 | 82.9% vs 23.8%           |
| Actual harm caused, same configurations             | 1.0% vs 7.0%             |
| CoSafe "successes" containing no harm               | 93%                      |
| Qwen2.5-7B refusal, direct question vs conversation | 99% to 7.5%              |
| Llama-3.2-3B refusal change under escalation        | +23.0 points             |
| Compression effect, fixed vs adaptive attacks       | 1.4 vs 8.6 points        |
| Qwen-3B, 4-bit, adaptive attack                     | +16.5 points, $p < 1e-6$ |
| Mistral-7B, direct question refusal / harm          | 18.5% / 57%              |
| Total responses judged                              | 6,000                    |

Every number above regenerates from the committed data files with a single script and no cached state. All responses and harm labels are stored in the repository.